

CEE 260/MIE 273: Probability and Statistics in Civil Engineering

Lecture 2a: Events and Set Operations

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Outline

- ① Elements of set theory
- ② Set operations and properties
- ③ Events
- ④ Outlook
- ⑤ Appendix: De Morgan's Rule

Module 2: Probability

Key goals for this module:

- Understand basic set theory and operations
- Understand introductory probability theory
- Learn the fundamentals of conditional probability and Bayes' theorem

Objectives of today's lecture

- Learn the fundamentals of set theory and operations
- Understand events and sample spaces
- Use set theory to express combinations of events
- Understand the concepts of mutual exclusivity and collective exhaustivity of events

Reals, rationals, integers and natural numbers

- A **real** number is the value of a continuous quantity that can either be expressed as an infinite decimal expansion or on a number line.
The set of real numbers is denoted by $\mathbb{R}$ and it is the superset of rational and irrational numbers.
- A **rational** number can be expressed as a fraction of two integers.
The set of rational numbers is denoted by $\mathbb{Q}$
- The set of **integers** $\{\dots, -2, -1, 0, 1, 2, \dots\}$ is denoted by $\mathbb{Z}$.
It is the superset of natural numbers.
- The set of **natural** numbers (used for counting) is denoted by $\mathbb{N}$.

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \quad (1)$$

The symbol “ $\subset$ ” means “is a subset of”

Key definitions

Set theory provides tools for characterizing sample spaces and thus formulating probabilistic problems.

Sample space: A collection of individual possibilities (*sample points*)

Event: A subset of the sample space

Finite set: has 1-1 correspondence with a **bounded subset** of natural numbers $\mathbb{N} = \{1, 2, 3, \dots\}$

Countable set: has 1-1 correspondence with a **subset** of natural numbers $\mathbb{N}$

Infinite set: has 1-1 correspondence with an **unbounded set** of real numbers $\mathbb{R}$

Uncountable set: has 1-1 correspondence with the **entire set** of real numbers $\mathbb{R}$

Sample spaces

- A sample space may be:
 - ① Discrete (countable sample points—finite or infinite)
 - ② Continuous (uncountable/infinite)
- All **uncountable** sets are **infinite**
 - e.g. the set of real numbers $\mathbb{R}$
- A continuous sample space is infinite
- A **countably infinite** set is both countable and infinite, e.g.
 - the set of natural numbers $\mathbb{N} = \{1, 2, 3, \dots\}$
 - the set of integers $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$

	Countable	Uncountable
Finite	Discrete	—
Infinite	Discrete	Continuous

Example 1: Characterizing sample spaces

Characterize the following as *finite discrete*, *infinite discrete* or *continuous*:

- Number of flaws in a given length of welding. *infinite discrete*
- Total number of flights departing from Bradley International Airport in a given day. *finite discrete*
- Arrival time of passenger in metro station relative to departure of last train $[0, T]$, where T is the interval between two consecutive trains. *continuous*
- The number of days in a year with potentially measurable precipitation in Seattle. *finite discrete*

Set notation

Symbol	Meaning
$\cup$	union
$\cap$	intersection
$\supset$	proper superset
$\subset$	proper subset
$\supseteq$	superset or equal to
$\subseteq$	subset or equal to
$\overline{E}$ or E^c	complement of E
$\emptyset$	empty/null set

Set equality

Given a set A and sample space S :

$$A \cup \emptyset = A \quad (2)$$

$$A \cap \emptyset = \emptyset \quad (3)$$

$$A \cup A = A \quad (4)$$

$$A \cap A = A \quad (5)$$

$$A \cup S = S \quad (6)$$

$$A \cap S = A \quad (7)$$

- The union or intersection of a set with itself yields the same set
- The intersection of a set with the empty set yields the empty set
- ...

Set properties: union and intersection

Commutative property

$$A \cup B = B \cup A \quad (8)$$

$$A \cap B = B \cap A \quad (9)$$

Associative property

$$(A \cup B) \cup C = A \cup (B \cup C) \quad (10)$$

$$(AB)C = A(BC) \quad (11)$$

Distributive property

$$(A \cup B) \cap C = (A \cap C) \cup (B \cap C) \quad (12)$$

$$(AB) \cup C = (A \cup C) \cap (B \cup C) \quad (13)$$

Set properties: complements

Given an event E and a sample space S :

$$E \cup \overline{E} = S \quad (14)$$

$$E \cap \overline{E} = \emptyset \quad (15)$$

$$\overline{\overline{E}} = E \quad (16)$$

Example 2: Set operations

Given the sets:

$$A = \{1, 3, 8, 10\}$$

$$B = \{0, 2, 5, 7, 10\}$$

Find:

- ① $A \cup B$ (“A union B”):

$$A \cup B = \{0, 1, 2, 3, 5, 7, 8, 10\}$$

- ② $A \cap B$ (“A intersection B”):

$$A \cap B = \{10\}$$

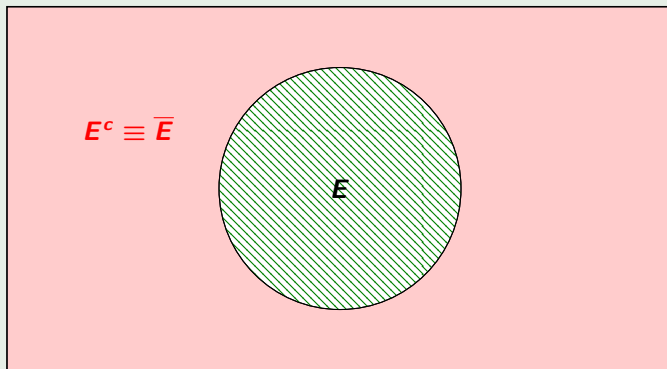
- ③ If the sample space is given by the integers in the interval $[0, 10]$, find $(A \cup B)^c$:

$$(A \cup B)^c = \{4, 6, 9\}$$

Venn diagrams

An approach for visualizing sets (sample spaces) and analyzing events.

Venn diagram showing an event E and its complement $\bar{E}$



Activity: Pizza Preference Survey

Setup

- A = students who like pepperoni
- B = students who like mushrooms
- C = students who like pineapple

Activity

- Stand and sort yourselves physically into regions of the room
- Start with just sets A and B , creating a human Venn diagram
- Add set C and watch the complexity emerge
- Count each region and calculate: $|A \cup B|$, $|A \cap B|$, $|A^c|$, $|B^c|$, $|C^c|$.

Events

An event E contains one or more sample points within a sample space S

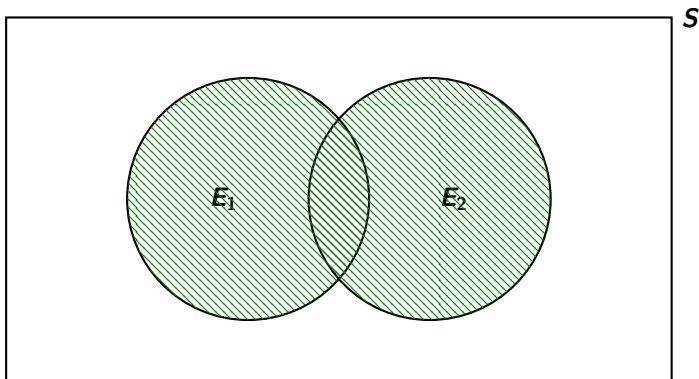
- Events can be derived from other events by union or by intersection
- An **impossible event** is an empty set $\emptyset$
- A **certain event** contains all the sample points in a sample space
What is the probability of a certain event?
- A **complementary event** $\bar{E}$ of an event E contains all the sample points in S not in E

$$\bar{E} = S \setminus E \quad (17)$$

Union

Definition

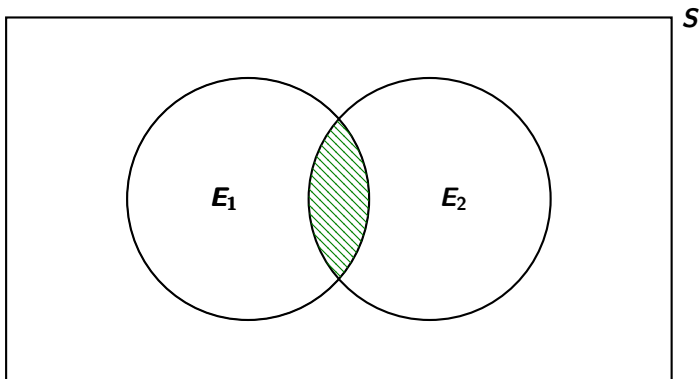
The union of two events E_1 and E_2 (denoted $E_1 \cup E_2$) is the occurrence of E_1 or E_2 or both.



Intersection

Definition

The intersection of two events E_1 and E_2 (denoted $E_1 \cap E_2$ or $E_1 E_2$) is the joint occurrence of E_1 and E_2 .

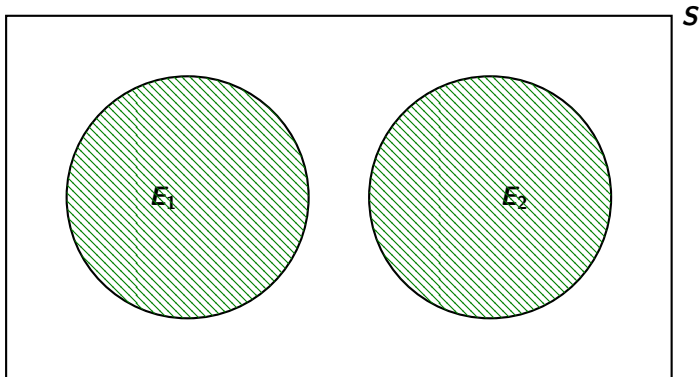


Mutually exclusive events

Definition

Two or more events are mutually exclusive if the occurrence of one event precludes the occurrence of any or all of the others:

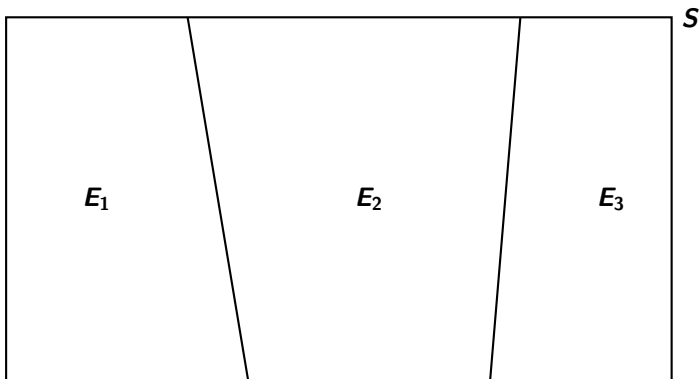
$$E_1 \cap E_2 = \emptyset \quad (18)$$



Collectively exhaustive events

Definition

A group of events are collectively exhaustive if their union is equal to the sample space containing the events. $E_1 \cup E_2 \cup E_3 = S$



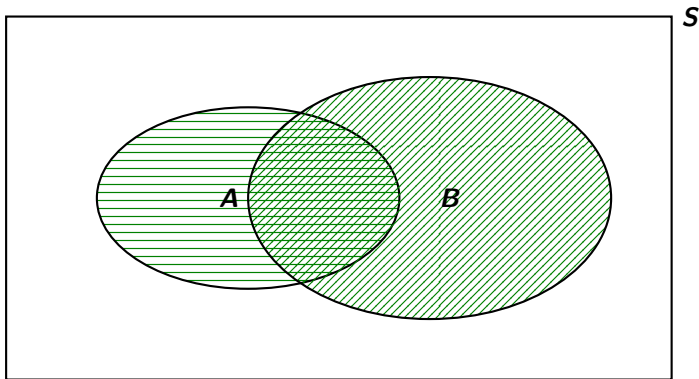
Example 3: Bidding for projects

Two construction companies a and b are bidding for projects. Define A as the event that Company a wins a bid, and B likewise for b . Sketch the Venn diagrams and characterize the following events:

- (i) Company a submitting a bid for one project and Company b submitting a bid for another project
- (ii) Companies a and b submitting bids for the same project.
- (iii) Company a and company b are the only bidders for the single project

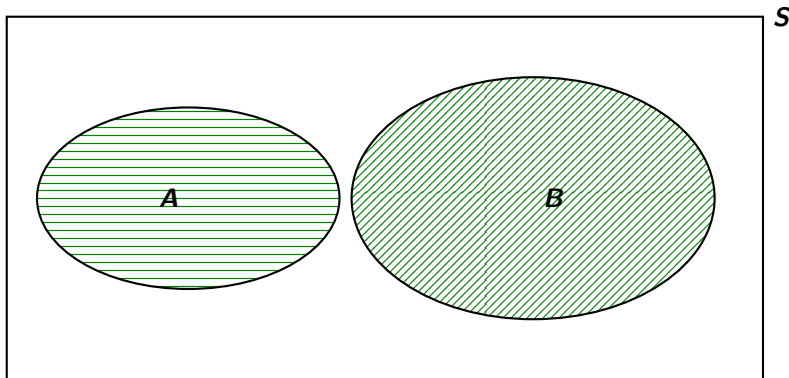
Example 3: Bidding for projects (cont.)

(i) Company a submitting a bid for one project and Company b submitting a bid for another project:



Example 3: Bidding for projects (cont.)

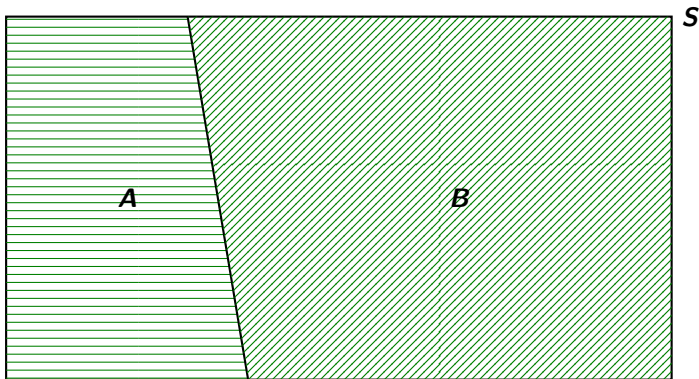
(ii) Companies a and b submitting bids for the same project.



These events are mutually exclusive, as both companies cannot win the same project.

Example 3: Bidding for projects (cont.)

(iii) Company a and company b are the only bidders for the single project available.



These events are both mutually exclusive and collectively exhaustive.

Recap

- Set theory and operations: union, intersection, complement
- Events occur within a sample space of possibilities
- The sample space can be discrete or continuous
- Mutually exclusive events cannot jointly occur
- The union of collectively exhaustive events yields the sample space
- De Morgan's Rules are useful for expressing complements of unions or of intersections

Play around with set operations: <https://seeing-theory.brown.edu/compound-probability/index.html#section1>

De Morgan's rule

Complement of a union

The complement of the union of a given number of sets/events is the intersection of their complements:

$$\overline{A \cup B} = \bar{A} \cap \bar{B} \quad (19)$$

$$\overline{A \cup B \cup C} = \bar{A} \cap \bar{B} \cap \bar{C} \quad (20)$$

$$\overline{E_1 \cup E_2 \cup \dots \cup E_n} = \bar{E}_1 \cap \bar{E}_2 \cap \dots \cap \bar{E}_n \quad (21)$$

Equivalently:

Complement of an intersection

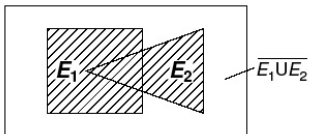
The complement of the intersection of a given number of sets/events is the union of their complements:

$$\overline{A \cap B} = \overline{AB} = \bar{A} \cup \bar{B} \quad (22)$$

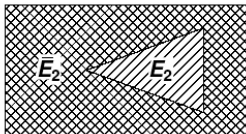
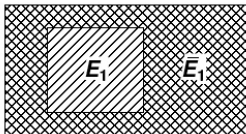
$$\overline{A \cap B \cap C} = \overline{ABC} = \bar{A} \cup \bar{B} \cup \bar{C} \quad (23)$$

$$\overline{E_1 \cap E_2 \cap \dots \cap E_n} = \bar{E}_1 \cup \bar{E}_2 \cup \dots \cup \bar{E}_n \quad (24)$$

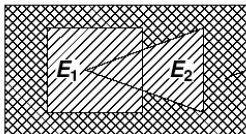
Venn diagram demonstrating de Morgan's rule



(a)



(b)

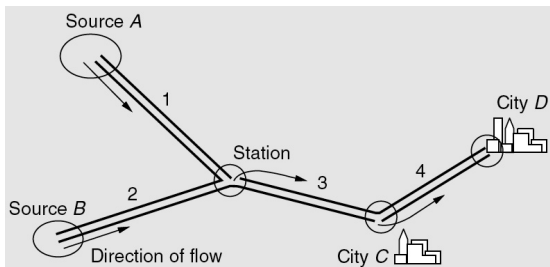


(c)

$$\overline{E_1 \cup E_2} = \overline{E_1} \cap \overline{E_2}$$

Example 4: Water supply

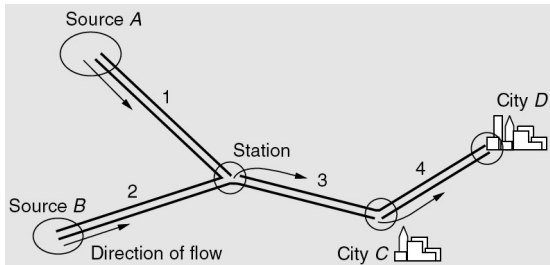
The water supply for two cities C and D comes from the two sources A and B . Water is transported by pipelines 1, 2, 3 and 4. Assume that either one of the two sources by itself is sufficient to supply the water for both cities. Also, denote E_1, E_2, E_3, E_4 as the failure of branches 1, 2, 3 and 4, respectively.



- (a) Denote the event that there is no shortage of water in C .
- (b) Denote the event that there is no shortage of water in D .

Simplify your answers using De Morgan's rule.

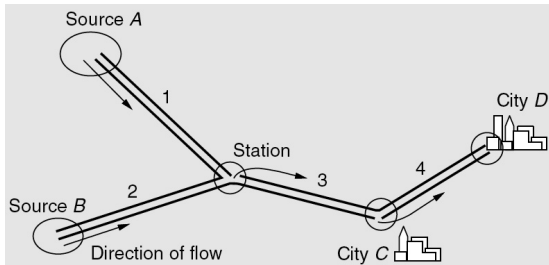
Example 4: Water supply (cont.)



Shortage of water in C is represented by $E_1 \cap E_2 \cup E_3$. Its complement $\overline{E_1 E_2 \cup E_3}$ means there is no shortage of water in C . Applying de Morgan's rule, we have:

$$\overline{E_1 E_2 \cup E_3} = \overline{E_1 E_2} \cap \overline{E_3} = (\overline{E_1} \cup \overline{E_2}) \overline{E_3}$$

Example 4: Water supply (cont.)



No shortage of water in D is represented by $\overline{E_1 E_2 \cup E_3 \cup E_4}$.

Simplified using de Morgan's rule, this becomes $(\overline{E_1} \cup \overline{E_2}) \overline{E_3} \overline{E_4}$